

Pouteria iWUE Analysis Report (Extended)

1. Factorial Model Results (iWUE ~ N * P)

Intercept (Control): 85.255

N effect: +14.543 (p = 0.0429)

P effect: +3.506 (p = 0.6359)

Interaction (N×P): -20.315 (p = 0.0665)

Derived NP effect vs control: -2.27

2. Treatment Model Results (iWUE ~ SUB + (1|Bloque))

Control: 25.326

N: +14.616 (p = 0.0429)

P: +3.524 (p = 0.6359)

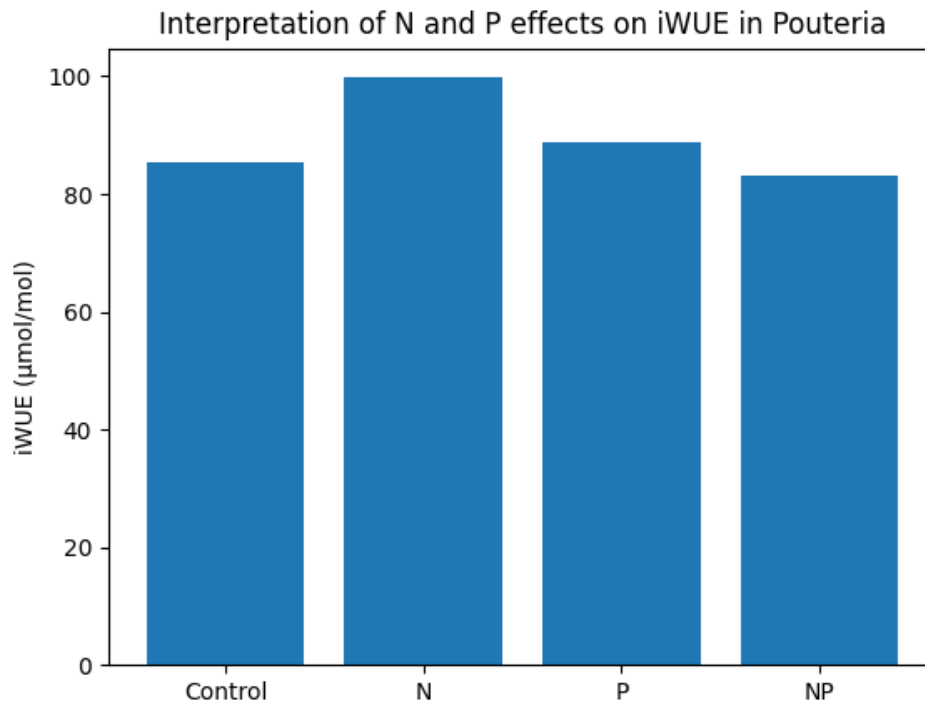
NP: -2.277 (p = 0.7685)

Random effect variance (Bloque): ~0

3. Interpretation

Nitrogen significantly increases iWUE, while phosphorus has no effect. The negative interaction indicates antagonism: phosphorus cancels the positive effect of nitrogen. The NP treatment is effectively similar to the control.

4. Interpretative Figure



5. Reporting Recommendation

Use both models: factorial (N × P) for mechanistic interpretation and treatment model for direct comparisons.

Suggested sentence:

Nitrogen addition significantly increased iWUE, while phosphorus alone had no effect. The combined treatment did not differ from the control, and the marginally significant negative interaction suggests an antagonistic response.